

Algorithmic Injustice: Sociotechnical Risks of Facial Recognition Technology

Biometric System Auditing & Tech Policy Analysis

Author: Hannah Rozenbaum, AI Safety & Risk Researcher

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Target System: Facial Recognition Technology

Executive Summary

Facial recognition technology (FRT) is a powerful identification method that has transformed security and identification systems. First developed in the 1960s by a team of computer scientists, the technology has significantly evolved through mathematical applications and increasingly vast datasets (NEC New Zealand, 2022). Despite notable improvements in technical accuracy, deployed models remain susceptible to demographic biases, raising serious socio-technical concerns related to privacy, consent, and civil rights.

The convenience and efficiency of tools like Face ID have normalized FRT as an everyday tool. However, the deployment of FRT in law enforcement and public surveillance poses significant risks of adverse consequences. Driven by market competition and tight corporate timelines, developers frequently prioritize commercial deployment over rigorous accuracy checks and comprehensive socio-technical risk assessments. This assessment evaluates the technical faults, corporate incentives, and social implications of imperfect FRT systems, arguing that unchecked deployment threatens fundamental human rights.

Technical Mechanism & Applications

FRT analyzes a face to identify an individual. Computer-generated filters, produced via artificial neural networks, convert faces into numerical data that can be compared and matched to an identity (Lewis & Crumpler, 2021). Gender, age, emotion, and other characteristics are used to classify an individual's face (Lewis & Crumpler, 2021). These traits are ambiguous yet laden with social, cultural, and political implications, making the process prone to bias.

FRT is applied in many fields, including law enforcement, social media, security, entertainment, device authentication, healthcare, and more. Facebook's tagging system and Amazon's facial recognition system (Rekognition) are well-known examples of how ubiquitous FRT has become. Given its cultural pervasiveness, the development and deployment of FRT should be carefully scrutinized.

Risk Analysis

FRT raises significant ethical and accuracy concerns. One key problem is the lack of consent when collecting facial data. Many people's faces are recorded and stored as data for policing purposes, yet they are often unaware that their data is being collected. Accessibility does

not equal ethics; as Boyd and Crawford (2012) argue, there is a key distinction between being in public and being public. A person unaware that their facial data is being collected, stored, and used has not consented, constituting a serious invasion of privacy. Many people are worried about how their data is used, with 81% of American adults reporting concern about how companies use the data they collect about them (McClain et al., 2023). Collecting facial data without consent is unethical and produces feelings of anxiety and worry, potentially leading to a loss of trust in government.

When used outside of a controlled environment in a real-world setting, FRT accuracy diminishes (Lewis & Crumpler, 2021). In real-world situations, when FRT misidentifies, it can lead to the misidentification of potential criminals and the wrongful conviction of innocent people. In the 2024 court case *Williams v. Detroit*, an innocent man was arrested due to an error in FRT (*Williams v. City of Detroit*, 2024). Since FRT is not entirely reliable, it is concerning that it is used to identify some criminals and arrest them with no other evidence. Misidentifying a criminal can ruin someone's life, potentially leading to wrongful arrest, reputational damage, emotional harm, disruption, and the financial burden of defending oneself in court. It is something that no one should endure, which is why FRT inaccuracy is so serious.

Diving Deeper: Biased Datasets & Their Harm

Issues of consent, privacy, and inaccuracy reflect the deeper biases in FRT. The datasets used to train FRT are not entirely representative. Bigger data is not always better data, as larger data cannot represent a whole population (Boyd & Crawford, 2012). A dataset lacking diversity is more likely to make errors. There are significant patterns in who is disproportionately impacted by misrepresentative training sets.

Amazon Rekognition has issues identifying darker skin tones and females, with only 68.6% accuracy for darker females (Buolamwini, 2019). A 2019 study tested photos of over eight million people and found that Asian and African Americans were 100 times more likely to be misidentified by FRT than white men ("You Can Opt out of TSA Face Scans," 2019). These misidentifications result from incomplete representation in model training. The misrepresentative training sets are formed by human decisions of which images to include. Humans' choice of faces can never be completely impartial, so biases infiltrate the technology. Artificial intelligence, including FRT, reflects eugenic ideals (Gebu & Torres, 2024). As training data scales in volume, it becomes more difficult to limit biases, resulting in discriminatory systems masked as innovative, beneficial technology.

Corporate Governance: Profit Over Public Welfare

Given these documented failures, critical questions arise regarding how companies address flaws in their FRT systems. Alarming, many choose to minimize or ignore them. This neglect is not an optimistic view of technology as a solution to social problems, but rather the result of corporate incentives driven by the expansion of capital and power. For instance, Amazon demonstrated little urgency in eliminating documented biases within the Rekognition software. The company neglected to submit its FRT systems to the National Institute of Standards and Technology facial recognition evaluations, and their claims of being bias-free were based on internal evaluations (Buolamwini, 2019). Their priorities lie in protecting their image, not fixing their technical flaws. Ultimately, computing is a story of capital. Corporations, driven by profit, invest in computing power and growth. Yet, this progress often comes at the expense of the safety, privacy, and civil liberties of marginalized communities who suffer most from algorithmic harm.

Case Study & Advocacy: UCLA Campus Resistance

The public has the power to prevent harm caused by FRT by understanding and denouncing its flaws. At the University of California Los Angeles (UCLA), students spoke out and shaped university policy. The UCLA administration was the first to propose implementing FRT for campus security. Students protested this decision, voicing the misidentification risks posed to marginalized communities (Burke, 2020). As a result of this mobilization, the university soon abandoned the proposal.

This case demonstrates the impact of public literacy and proactive intervention in curbing inequitable FRT practices. It is necessary to critically investigate the development and usage of such technology, ensuring it upholds and protects rights to privacy, freedom, and equality.

Strategic Recommendations

To mitigate the socio-technical risks associated with FRT, policymakers should implement preventative measures. During the training process, data governance bodies should establish rigid, written guidelines for training data collection and curation. These rules should include diverse sourcing, frequent dataset updates and monitoring, and external human review. Before deploying new software to the public or using software for a new application, independent, third party auditing should be mandatory. Internal self evaluations should not be

relied upon as the sole pre-deployment check. Once software has passed third party screening, it should be released with explicit documentation and consent procedures. Anyone subject to facial data collection should be informed and be able to opt out. In criminal law, FRT should not be the sole cause for arrest, preventing wrongful convictions like *Williams v. Detroit*.

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